

Vivienne Chow

Software Engineer · San Francisco, CA

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SUMMARY

Backend and infrastructure engineer working at the intersection of large-scale systems and AI. Currently building safety and reliability infrastructure behind Google's Gemini models. Previously worked on some of the highest-traffic pages on the internet at Amazon.

EXPERIENCE

Google — Software Engineer, Systems & AI Infrastructure

Oct 2022 – Present

- Build safety and reliability infrastructure for Google Gemini, keeping generation safe and fast for Vertex AI, Google AI Studio, and consumer apps within a strict <50ms latency budget.
- Led a multi-quarter redesign of a large-scale distributed database for multi-region resilience, removing a whole class of outages.
- Built emergency override systems that cut incident resolution time from hours to minutes during high-stakes regulatory rollouts across 18 teams' systems.
- Built monitoring that raised detection of silent data-quality issues from near zero to over 95%.

Amazon — Software Development Engineer

Jun 2021 – Sep 2022

- Performance and reliability work on Amazon's product detail pages — one of the highest-traffic pages on the internet — improving load speed for millions of daily visitors.

ConovaAI — Cloud Developer

May 2020 – May 2021

- Built serverless data pipelines and ETL infrastructure.

SKILLS

Languages Python · Go · Java · TypeScript · C++ · SQL

Data structures & algorithms Trees · Graphs · Hash maps · Heaps · Tries · LRU caches · Dynamic programming · Concurrency primitives

Systems & infrastructure Distributed systems · Multi-region failover · Kubernetes · gRPC · Event-driven architecture · Low-latency serving · Capacity planning

Cloud & data GCP · AWS · Spanner / BigQuery · Kafka · Terraform · CI/CD

AI safety & evaluation LLM safety infrastructure · Policy & guardrail systems · Red-teaming · Adversarial testing · Evaluation harnesses & benchmarking · Content moderation pipelines · Responsible AI tooling

AI infrastructure & inference LLM serving at scale · Inference optimization (batching, KV cache, quantization) · GPU/TPU accelerator scheduling · RAG · Vector databases · MLOps · Model rollout & canarying

Reliability SRE practices · SLOs & error budgets · Observability & tracing · Incident response & postmortems · Chaos & load testing · Performance tuning · Large-scale system design